C 24785 (Pages : 3) Name.....

Reg. No.....

SECOND SEMESTER B.Sc. DEGREE EXAMINATION, MAY 2017

(CUCBCSS—UG)

Complementary Course

STS 2C 02—PROBABILITY DISTRIBUTION

Time : Three Hours Maximum : 80 Marks

Section A

Answer **all** *questions in* **one word.**

Each question carries 1 *mark.*

Name the following :

1. For a random variable X , the function $E\left[e^{itX}\right]$.
2. For two random variables X and Y, a function denoting $P(X = x, Y = y)$.
3. If $X \sim N(\mu, \sigma)$, then the distribution of the square of X.

Fill up the blanks :

4. Correlation between two random variables X and Y is $\frac{1}{4}$, Cov (X, Y) is $\frac{1}{36}$, then the product of standard deviations of X and Y is ————.
5. X is a Poisson random variable with $E(X) = 5$. $E\left(X^2\right)$ is ————.
6. X is the number shown when a fair die is tossed, X follows ———— distribution.
7. If $X \sim N(4, 2)$, then $P(X > 4) =$ ————.

Write True *or* False :

8. Second cumulant k_2 of a random variable X is V (X).
9. The mean of a Binomial random variable is less than its variance.
10. In some situations Tchebycheve's inequality may give a negative lower bound.

(10 × 1 = 10 marks)

Turn over

Section B

Answer **all** *questions in* **one sentence** *each.*
Each question carries 2 *marks.*

11. State any *four* properties of expectation.
12. Obtain the expected number of heads obtained in three tosses of a fair coin.
13. Define conditional expectation.
14. $X \sim N(\mu_1, \sigma_1)$ and $Y \sim N(\mu_2, \sigma_2)$, obtain the distribution of $Y = aX + bY$, where X and Y are independent.
15. Define rectangular distribution
16. Define Pareto distribution.
17. State Central Limit theorem.

(7 × 2 = 14 marks)

Section C

Answer any **three** *questions.*
Each question carries 4 *marks.*

18. Given the pdf of X, $f(x) = k\left(1 - x^2\right), 0 < x < 1$. Prove that $E(X) = \frac{3}{8}$.
19. Obtain the mgf of the random variable X with p.d.f. $f(x) = \frac{1}{2}e^{-|x|}, -\infty < x < \infty$.
20. Obtain the mgf of Poisson random variable with a parameter λ.
21. If X follows gamma distribution with one parameter p, obtain V(X).
22. Define convergence in probability and convergence in distribution.

(3 × 4 = 12 marks)

Section D

Answer any **four** *questions.*
Each question carries 6 *marks.*

23. For two random variables X and Y, the joint p.d.f.

$f(x, y) = 2 - x - y ; 0 \le x \le 1, 0 \le y \le 1$
$= 0$, otherwise

. Find Cov (X, Y).

24. State and prove Cauchy-Schwartz inequality. Use it to prove that $-1 \le r_{xy} \le 1$, where r_{xy} is Pearson's coefficient of correlation between any two random variables X and Y.

25. State and prove the lack of memory property of exponential distribution.

26. Prove that $\mu_{r+1} = pq\left[\frac{d}{dp}\mu_r + nr\mu_{r-1}\right]$, where $\mu_{r-1}, \mu_r, \mu_{r+1}$ are the central moments of binomial distribution with parameters n and p.

27. If $X \sim N(\mu, \sigma)$, show that the mean deviation about the mean is $\sqrt{\frac{2}{\pi}}\sigma$.

28. State and prove Tchebycheve's inequality.

(4 × 6 = 24 marks)

Section E

Answer any **two** *questions.*
Each question carries 10 *marks.*

29. Let X and Y are two random variables with joint pmf $f(x, y) = \frac{x+y}{21}$ $x = 1, 2, 3$; $y = 1, 2$.

Find :

(i) Correlation between X and Y

(ii) $V(X/Y = 2)$.

30. (i) For a geometric random variable with parameter p, obtain mean, variance and m.g.f.

(ii) If X and Y are two independent geometric random variables with same parameter p, show that the conditional distribution of X given $X + Y = z$ follows uniform distribution.

31. (i) Define beta distribution of first and second kind. Obtain their means.

(ii) If X follow beta distribution of first kind with parameters p and q, show that $Y = X/(1 - X)$ follow beta distribution of second kind.

32. State and prove Weak law of large numbers. For a sequence of random variables (X_k). Given that $P(X_k = -2^k) = 2^{-(2k+1)} = P(X_k = 2^k)$, $P(X_k = 0) = 1 - 2^{-(2k+1)}$, Examine if the law of large numbers holds for this sequence.

(2 × 10 = 20 marks)